Prana CO2 Sensor Calibration Protocol

Goal

Run each ESP32 + Prana Air CO2 node beside an **Aeroqual S500 with its CO2 head** for 2–3 days. Average both to 1 minute, then fit a **linear regression** so the Prana reads like the Aeroqual.

The method follows Bose et al. [1], who ran Prana sensors beside an Aeroqual S-500 for 20 days at one reading per minute, with the windows closed. They found Prana needs local calibration even though it is factory calibrated. Our run is a shorter 2–3 day pilot, with the node logging every 15 s.

Equipment

Item	Purpose	Check before starting
Aeroqual S500 + **CO2 head**	Reference instrument, as in [1]	Correct head fitted; calibration status of the head; clock set; log cleared
ESP32 + Prana Air CO2 nodes	Sensors to calibrate	Each node has a unique ID; RTC synced by NTP; SD card inserted
Clean 5 V power supply for every node	Stable NDIR readings	Use a supply already proven to read correctly (a noisy adapter shifted readings to ~2000 ppm)
Wi-Fi hotspot / router	NTP time sync and ThingSpeak upload	Nodes connect and post before the run starts
Laptop	Download Aeroqual log and SD card data	Aeroqual software installed

One S500 = one head at a time. The S500 takes one sensor head at a time, so the CO2 run cannot share an S500 with a PM calibration run. Run CO2 as its own session with the CO2 head fitted, or use a second S500.

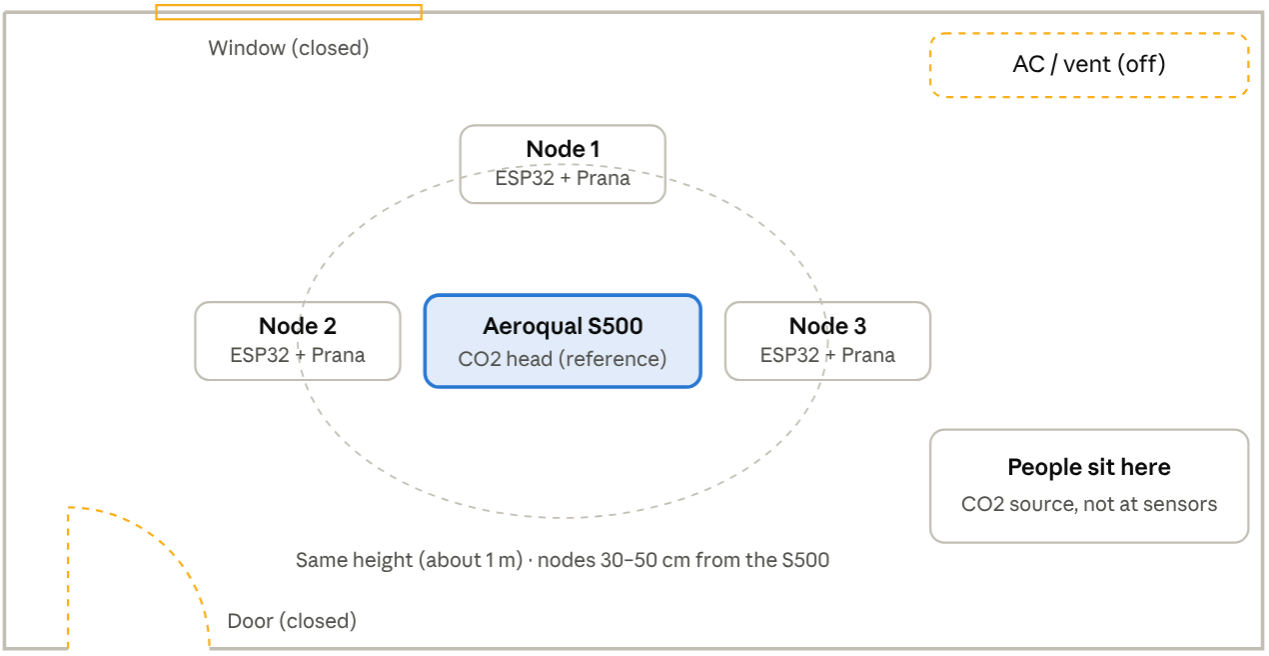
No smoke or incense in this session. Burning adds CO2 near the source and makes the room uneven, so the paired readings stop sampling the same air.

Room placement

Put every node right beside the Aeroqual so they all breathe the same air. Keep doors and windows closed and the AC or fans off during the run, as the paper did.

All nodes sit beside the Aeroqual, away from vents, windows and breath

Top view of the calibration room, not to scale



Orange = keep sensors away. Also avoid breath, heat sources, and any smoke or incense.

calibration room layout · top view, not to scale

With only one node, place it directly beside the S500. Give every node a unique ID label before the run.

Data collection plan

To reach **20,000 node readings at 15 s, run for about 3.5 days (84 hours)**. Three days gives 17,280 readings. The Aeroqual logs once a minute, so the number of **paired** rows for training is set by the Aeroqual, not by the node.

Duration	Node readings (every 15 s)	Aeroqual readings (every 1 min) = paired rows
1 hour	240	60
1 day	5,760	1,440
2 days	11,520	2,880
3 days	17,280	4,320

Duration	Node readings (every 15 s)	Aeroqual readings (every 1 min) = paired rows
3.5 days	20,160	5,040

After collection, the four node readings in each minute are averaged into one value, so 20,000 node readings become about 5,000 paired 1-minute rows.

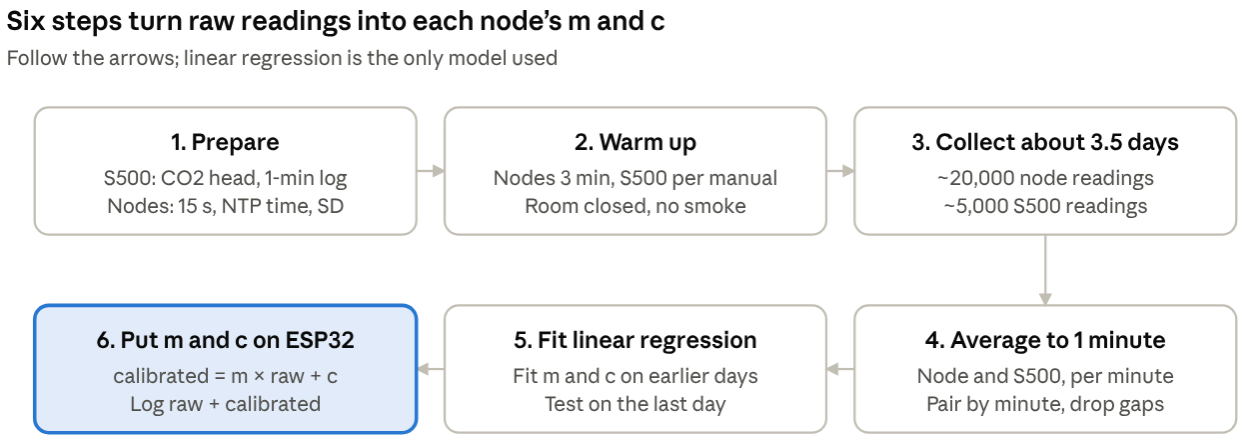
Settings:

- Node firmware: `SAMPLE_INTERVAL_MS = 15000`. The SD card keeps every reading; ThingSpeak is only a live view.
- Aeroqual S500: logging interval **1 minute**, clock set to the same time zone as the nodes (IST).
- Log raw Prana ppm only. Do not apply any correction while collecting.

Cover a wide CO2 range. A correction is only valid inside the range it was trained on. The paper saw about 400–2100 ppm. Include quiet periods (empty room, night, near 400–500 ppm) and occupied periods with the door and windows closed, so CO2 rises. Write down when people enter and leave, and when doors or windows open.

Procedure at a glance

The whole calibration is six steps, ending with each node's own m and c on its ESP32.



calibration procedure · 6 steps

Do not start step 3 until every node has passed its warm-up and is logging to its SD card.

Processing the data

1. **Back up** the raw SD card CSV and the Aeroqual log before changing anything.

2. **Average both to 1 minute.** Take the mean of the node readings in each minute (up to 4 at 15 s), and the mean of the Aeroqual readings in each minute. Keep only node rows with `time_valid = 1`.
3. **Pair by minute.** Match each 1-minute Prana value with the 1-minute Aeroqual value for the same minute. Drop minutes where either side is missing.
4. **Remove outliers** with a Z-score above 3, as the paper did. If the Aeroqual shows the same peak, keep it: it is a real event.
5. **Split by day.** Use the earlier days to fit the model and the last day to test it (e.g. days 1–2 fit, day 3 test).

Linear regression

Fit one straight line from the 1-minute Prana value (x) to the 1-minute Aeroqual value (y). This is the linear regression from the paper:

```
\text{CO2}_{\text{calibrated}} = m \times \text{CO2}_{\text{Prana raw}} + c
```

Fit m and c on the training days, then test on the last day:

```
import numpy as np

# train / test = paired 1-minute rows with columns prana_ppm, aeroqual_ppm
m, c = np.polyfit(train.prana_ppm, train.aeroqual_ppm, 1)

pred = m * test.prana_ppm + c
rmse = np.sqrt(np.mean((pred - test.aeroqual_ppm) ** 2))
r2 = 1 - np.sum((test.aeroqual_ppm - pred) ** 2) / np.sum((test.aeroqual_ppm
- test.aeroqual_ppm.mean()) ** 2)

print(f"m = {m:.4f}, c = {c:.2f}, R2 = {r2:.3f}, RMSE = {rmse:.1f} ppm")
```

What to expect, from the paper: with linear regression, Prana reached $R^2 = 0.796$ at 1-minute averages. Its error dropped from 24.68 before calibration to 5.02 after (1-hour averages, the paper's Erms).

Putting the correction on the ESP32

Each node gets **its own** correction, trained on its own Prana unit. Never copy one node's numbers to another node.

Add the node's m and c to its firmware as two constants and one line:

```
// Calibration for node AQ-01 (its own Prana unit), trained on <dates>
const float CO2_SLOPE     = 0.0;   // m from training
const float CO2_INTERCEPT = 0.0;   // c from training
const int   CO2_TRAIN_MIN = 400;   // lowest raw ppm seen in training
const int   CO2_TRAIN_MAX = 2100;  // highest raw ppm seen in training

int  co2Calibrated = (int)(CO2_SLOPE * co2Raw + CO2_INTERCEPT + 0.5);
bool co2InRange    = (co2Raw >= CO2_TRAIN_MIN && co2Raw <= CO2_TRAIN_MAX);
```

Always:

- Log **both** the raw and the calibrated value to the SD card, so the correction can be checked or replaced later.
- Record the node ID, Prana unit, training dates and model type next to the constants.
- Flag readings outside the trained range (`co2InRange = false`); the correction was never tested there.

Final checklist

- [ ] S500 fitted with the CO2 head, logging every 1 minute, clock set to IST
- [ ] Nodes on clean 5 V supplies, logging raw Prana ppm every 15 s to SD, each with a unique ID
- [ ] All sensors warmed up; room closed; no smoke or incense
- [ ] Nodes beside the S500 at the same height, away from vents, windows, doors and breath
- [ ] Ran about 3.5 days (at least 2–3 days)
- [ ] Raw data backed up
- [ ] Both averaged to 1 minute and paired by minute; Z-score outliers checked
- [ ] Linear regression fitted on earlier days, tested on the last day
- [ ] Each node's m and c stored with its node ID; raw values still logged

Reference

[1] R. Bose, A. Parmar, H. Narla and S. Chaudhari, "Comparative Evaluation of Low-Cost CO2 Sensors for Indoor Air Pollution Monitoring," *2022 IEEE 8th World Forum on Internet of Things (WF-IoT)*, 2022, doi: 10.1109/WF-IoT54382.2022.10152133. IEEE Xplore · Full text (IIIT Hyderabad)

All paper figures in this doc come from the full text above. The 15 s node rate, the 2–3 day run and the day-based test split are our choices, not the paper's.